4 Number Sense

Big idea. "Number sense" is the ability to *understand* numbers, not just to *compute* with them: knowing which number is larger, whether a result is reasonable, how to round for a quick estimate, and seeing the hidden structure inside a number (its digits, its divisors, its multiples). In this topic you will review and connect:

- **decimals and negative numbers**: comparing and ordering them on the number line;
- the **order of operations** and **estimating** results;
- **fractions, ratios, percentages**: "a fraction of a number", equivalent fractions, discounts;
- **speed** – distance – time and unit conversion (km/h, m/s);
- **divisibility, common divisors, last digits, place value** (solving digit puzzles);
- the **mean, the median** and **number patterns** (Fibonacci, arithmetic sequences).

The general secret: *estimate before you calculate; check after you calculate.* A student with good number sense can often eliminate several unreasonable options in a multiple-choice question without any exact calculation.

① Decimals: reading, comparing, ordering, adding and subtracting

1. Place value. In a decimal, each digit sits in a *place*. After the decimal point come the tenths, the hundredths, the thousandths, ...

Tens	Ones	.	Tenths	Hundredths	Thousandths
	2	.	2	3	
	2	.	3	0	

Example: $2.23 = 2 + \frac{2}{10} + \frac{3}{100}$, while $2.3 = 2 + \frac{3}{10} = 2.30$.

2. Adding or removing zeros at the end of the decimal part does not change the value: $2.3 = 2.30 = 2.300$. Reason: $\frac{3}{10} = \frac{30}{100} = \frac{300}{1000}$ (multiply numerator and denominator by 10).

3. Rule for comparing two positive decimals.

- Compare the *whole-number parts* first: the one with the larger whole part is larger.
- If the whole parts are equal, *add zeros* so that both decimal parts have the same number of digits, then compare place by place from left to right: tenths, then hundredths, ...

Example: compare 0.304 and 0.34. Write $0.34 = 0.340$. Tenths: $3 = 3$; hundredths: $0 < 4$. So $0.304 < 0.340 = 0.34$.

4. Adding and subtracting decimals: write the numbers in a column *with the decimal points lined up* (places lined up), then add/subtract as with whole numbers.

5. Rounding decimals. To round to the nearest hundredth (two digits after the point), look at the digit just to the right of that place (the thousandths digit): if it is less than 5, keep the

hundredths digit; if it is 5 or more, add 1 to the hundredths digit; then drop all digits after it. Example: $4.6875 \approx 4.69$ (the thousandths digit is $7 \geq 5$), while $3.1416 \approx 3.14$ (because $1 < 5$). The sign "$\approx$" reads "is approximately".

Why is it true? Why can't we compare "by length" (the number with more digits is larger)? Write both in thousandths: $0.304 = \frac{304}{1000}$ and $0.34 = \frac{340}{1000}$. The fractions have the same denominator and $304 < 340$, so $0.304 < 0.34$, even though 0.304 "looks longer". Adding zeros is exactly bringing both to the same denominator. □

Common mistake. Many students think $2.23 > 2.3$ because "$23 > 3$". Wrong! Compare place by place: 2.23 has 2 tenths, while 2.3 has 3 tenths, so $2.23 < 2.3$.

ThinkAlgo Tip. To find the smallest (largest) number in a set of decimals, write all of them in *one column* with the decimal points lined up, adding zeros to make them equally long. Then you just compare them like whole numbers.

② Negative numbers and the number line

1. The number line. The integers are placed on a straight line: 0 in the middle, positive numbers to the right, negative numbers to the left.



2. Comparison rule. On a (horizontal) number line, *a number to the left is smaller than a number to the right.* Therefore:

- every negative number is less than 0, and 0 is less than every positive number;
- every negative number is less than every positive number;
- of two negative numbers, the one *farther from* 0 is *smaller*: $-8 < -2$ because $8 > 2$.

3. Everyday pictures. A temperature of $-8°C$ is colder than $-2°C$; owing 8 thousand dong leaves you "poorer" than owing 2 thousand dong.

4. The coordinate plane. A point is given by a pair of numbers $(x; y)$: x is the horizontal coordinate (moving right/left), y is the vertical coordinate (moving up/down). A horizontal move changes only x; a vertical move changes only y.

Why is it true? Why is $-8 < -2$? We have $-2 = -8 + 6$: to go from -8 to -2 on the number line we must move 6 units *to the right*. Moving right means increasing, so -2 is larger than -8. □

Common mistake. Do not compare negative numbers "by the feel of positive numbers". -8 is *less* than -2, even though 8 is greater than 2. When in doubt, sketch a quick number line.

ThinkAlgo Tip. **Parity (even − odd)** is a powerful "number sense" tool: even + even = even, odd + odd = even, even + odd = odd; a product of odd numbers is always odd. For instance, the

sum of 5 odd numbers is odd, and the sum of 4 odd numbers is even. When a point moves on a coordinate grid, looking at the parity of its coordinates often tells you at once whether a position can be reached.

③ Order of operations and estimation

1. Order of operations.

1. *Brackets* first: ( ), then [ ], then { }.
2. Next, *powers*.
3. Then *multiplication and division*, from left to right.
4. Finally *addition and subtraction*, from left to right.

Example: $6\times2-3=12-3=9$ (multiply first, then subtract), whereas $6\times(2-3)=6\times(-1)=-6$.
2. Rounding to estimate. To know quickly "roughly how much" a result is, round the factors to the nearest ten or hundred and compute mentally. Example: $99\times9\approx100\times9=900$.
3. A quick-calculation trick: $99\times9=(100-1)\times9=900-9=891$ (multiplication distributes over subtraction).

Why is it true? Why is $(100-1)\times9=100\times9-1\times9$? Imagine 9 rows of 100 sweets each, but 1 sweet is taken from every row. The sweets left are 9 rows × 99 sweets, which also equals (9 × 100 sweets) minus (the 9 sweets taken). This is the *distributive property*: $a\times(b-c)=a\times b-a\times c$.□

Common mistake. Working "from left to right regardless of the operation" is wrong: $2+3\times4\neq5\times4$. The correct value is $2+3\times4=2+12=14$.

ThinkAlgo Tip. In a multiple-choice question asking "which number is closest to…", the options often differ by factors of 10 (10; 100; 1000; …). Then a rough estimate is enough to choose correctly.

④ Fractions: a fraction of a number, equivalent fractions

1. A fraction is a part of a whole. $\frac{a}{b}$ means: split the whole into b equal parts and take a of them.
2. Finding a fraction of a number: to find $\frac{m}{n}$ of a, compute $a\times\frac{m}{n}$ (split a into n parts and take m of them). Example: $\frac{1}{3}$ of 396 is $396\div3=132$.
3. The part that is left. If $\frac{m}{n}$ of a job is done, the part left is $1-\frac{m}{n}=\frac{n-m}{n}$.
4. Equivalent fractions (the basic property). Multiplying (or dividing) the numerator and the denominator by the same non-zero number gives an equal fraction:

$$\frac{3}{7}=\frac{3\times9}{7\times9}=\frac{27}{63},\qquad\frac{12}{36}=\frac{12\div12}{36\div12}=\frac{1}{3}.$$

5. Simplifying a fraction: divide the numerator and the denominator by their greatest common

divisor to get a fraction in lowest terms.

6. "Sum and ratio" problems. If two numbers are in the ratio $\frac{5}{8}$, we can think of the first number as 5 equal parts and the second as 8 such parts. Knowing the sum (or the difference) gives the value of one part.

7. A fraction of a fraction. "Half of a quarter" is $\frac{1}{2} \times \frac{1}{4} = \frac{1}{8}$: multiply numerator by numerator, denominator by denominator. This happens often when a shape is halved several times in a row: each halving multiplies the area fraction by $\frac{1}{2}$ again.

8. Comparing fractions. Bring them to a common denominator and compare numerators, or convert them to decimals. Examples: $\frac{2}{3} = \frac{8}{12} > \frac{7}{12}$; $\frac{3}{5} = 0.6 < \frac{3}{4} = 0.75$.



Why is it true? Why does multiplying the numerator and denominator by the same number leave the fraction unchanged? $\frac{3}{7}$ is "cut the cake into 7 parts, take 3". If we cut *each* part into 9 smaller pieces, the cake has $7 \times 9 = 63$ pieces and our share has $3 \times 9 = 27$ pieces. The amount of cake has not changed, so $\frac{3}{7} = \frac{27}{63}$. □

Common mistake. Read the question carefully: does it ask for the part *done* or the part *left*? Many students correctly find $\frac{1}{3}$ of 396 to be 132 and circle it straight away, while the question asks for the pages *left*, $396 - 132 = 264$.

Note for parents. With fractions, a picture beats any explanation. Draw a strip (a chocolate bar) split into equal parts and shade the part used; your child will see the part left by themselves. This is the familiar "bar model" from primary school.

⑤ Percentages, unit price and discounts

1. Percent means "out of one hundred". $80\% = \frac{80}{100} = 0.8$.

2. Finding $p\%$ of a number a: compute $a \times \frac{p}{100}$. Example: 80% of 50 is $50 \times \frac{80}{100} = 40$.

3. Expressing a as a percentage of b: compute $\frac{a}{b} \times 100\%$. Example: 32 correct answers out of 40 questions is $\frac{32}{40} \times 100\% = 80\%$.

4. Unit price. The unit price is the price of *one* unit (one kilogram, one book, ...):

$$\text{Total cost} = \text{unit price} \times \text{quantity}, \qquad \text{unit price} = \text{total cost} \div \text{quantity}.$$

5. A discount of $p\%$. New price = old price $\times \left(1 - \frac{p}{100}\right)$. A 10% discount leaves 90%: multiply by 0.9.

6. Successive discounts. Each time the price is reduced by 10% *of the current price*, we multiply by 0.9 once more. Example: a price of 100 thousand dong after two reductions: $100 \to 90 \to 81$ (thousand dong), *not* 80.

Why is it true? Why are two 10% reductions not the same as one 20% reduction? The second reduction is 10% of 90 (that is 9), not 10% of 100 (that is 10). Since the price has already dropped, the next reduction is smaller too: in total the price drops by only $10 + 9 = 19$ thousand dong, i.e. 19%. □

ThinkAlgo Tip. Finding 10% very fast: just divide by 10 (move the decimal point one place to the left). From 10% you get 5% (halve it), 20% (double it), $80\% = 100\% - 20\%$, ...

⑥ Speed, distance, time and unit conversion

1. Three linked formulas. For motion at a constant speed:

$$s = v \times t, \qquad v = s \div t, \qquad t = s \div v,$$

where s is the distance, v the speed and t the time.

2. Who is faster? The faster person covers a *longer* distance in the *same amount of time*. The safest way: compute each person's speed in the same unit (for example m/s) and compare.

3. Running laps. Running n laps of a track L metres long means a distance of $n \times L$ metres.

4. Converting km/h to m/s. 1 km = 1000 m and 1 hour = 3600 seconds, so

$$1 \text{ km/h} = \frac{1000 \text{ m}}{3600 \text{ s}} = \frac{5}{18} \text{ m/s}, \qquad 18 \text{ km/h} = 18 \times \frac{5}{18} = 5 \text{ m/s}, \qquad 1 \text{ m/s} = 3.6 \text{ km/h}.$$

$s = v \times t$

Cover the quantity you want:
$v = s \div t$ and $t = s \div v$

Common mistake. The quantities must use *matching units*: if the distance is in metres and the time in seconds, the speed comes out in m/s; do not divide metres by hours. Before comparing two speeds, convert them to the same unit.

⑦ Divisibility, common divisors, last digits and place value

1. Multiples and divisors. a is a *multiple* of b (or b is a *divisor* of a) if $a = b \times k$ for some whole number k. Example: $45 = 9 \times 5$, so 45 is a multiple of 9.

2. Divisibility rules.

- by 2: the last digit is $0, 2, 4, 6, 8$; by 5: the last digit is 0 or 5; by 10: the last digit is 0;
- by 3: the digit sum is divisible by 3; by 9: *the digit sum is divisible by* 9.

3. Common divisors. A number that divides both a and b is a common divisor of a and b. Every

common divisor is a divisor of the *greatest common divisor* $\gcd(a, b)$. Example: $\gcd(40, 72) = 8$, so the common divisors of 40 and 72 are $1, 2, 4, 8$.

4. The last digit of a product depends only on the last digits of the factors: $\ldots 3 \times \ldots 7$ ends in the same digit as $3 \times 7 = 21$, namely 1.

5. Place value. A number written with digits can be expanded by place:

$$\overline{abc} = 100a + 10b + c, \qquad \overline{PQQ} = 100P + 10Q + Q = 100P + 11Q.$$

(The bar on top shows that this is the number with digits a, b, c, not the product $a \times b \times c$.)

Why is it true? Why does the divisibility rule for 9 work? Take the number $\overline{abc} = 100a+10b+c$ and split it:

$$100a + 10b + c = (99a + 9b) + (a + b + c) = 9 \times (11a + b) + (a + b + c).$$

The part $9 \times (11a + b)$ is always divisible by 9. So $\overline{abc}$ is divisible by 9 exactly when the remaining part $a + b + c$ (the digit sum) is divisible by 9. For numbers with more digits the argument is the same, because $10 = 9 + 1$, $100 = 99 + 1$, $1000 = 999 + 1$, $\ldots$ □

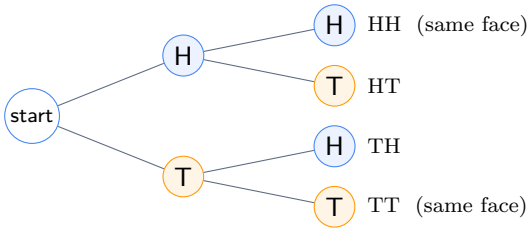
ThinkAlgo Tip. For a "digit puzzle" (each letter stands for a digit) such as $\overline{PQQ} + \overline{PPQ} + \overline{QQQ} = 876$, work *from the ones column to the left*, watching out for carries. The ones column usually has the fewest possibilities, so the first digit is found very quickly. You can also use place value to turn the puzzle into one equation. When you have to count or find all possibilities (sets of digits, outcomes of tossing coins, $\ldots$), *list them systematically* (in increasing order, or with a table/tree diagram) so that nothing is missed or repeated.

Systematic listing and chance.

- To count the three-digit numbers whose digits have a given *product*, find the sets of three digits (written in increasing order so as not to repeat), then count the arrangements of each set: three different digits can be arranged in $3 \times 2 \times 1 = 6$ ways; two equal digits and one other in 3 ways. Example: product 12 with the set $\{1; 3; 4\}$ gives 6 numbers $134, 143, 314, 341, 413, 431$; the set $\{2; 2; 3\}$ gives 3 numbers $223, 232, 322$.
- When all outcomes are *equally likely*, the probability of an event is

$$\frac{\text{number of favourable outcomes}}{\text{total number of possible outcomes}}.$$

Example: tossing 2 coins has 4 equally likely outcomes: HH, HT, TH, TT (H: heads, T: tails). The probability that both coins show the same face is $\frac{2}{4} = \frac{1}{2}$. Use a tree diagram to list them:



Common mistake. "The digit sum is divisible by 9" is the rule for 9; it does *not* work for 4, 6 or 8. And a number divisible by 3 is not necessarily divisible by 9: 12 is divisible by 3 but not by 9.

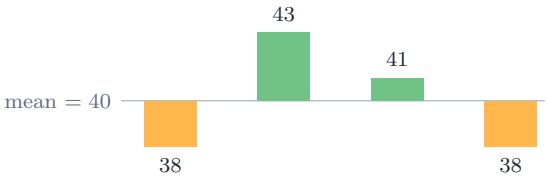
⑧ Number patterns, the mean and the median

1. The Fibonacci sequence. $1, 1, 2, 3, 5, 8, 13, 21, 34, 55, \ldots$ Each number (from the third on) is *the sum of the two numbers just before it*: $2 = 1 + 1$, $3 = 1 + 2$, $5 = 2 + 3$, ...
2. Arithmetic sequences (equally spaced). Each number (from the second on) equals the previous number plus a fixed number d (the common difference). Example: $1, 4, 7, 10$ (with $d = 3$). The n-th number is $a_1 + (n - 1) \times d$.
3. The mean (average). The mean of n numbers = (their sum) $\div n$. Hence

$$\text{Sum} = \text{mean} \times \text{how many numbers.}$$

4. The "deviation" method. The deviations from the mean always add up to 0: the "excess" of the numbers above the mean exactly equals the "shortfall" of the numbers below the mean.
5. The median. Arrange the numbers in increasing order; the number in the middle is the median (when there is an odd number of numbers). Example: the median of $42, 43, 46, 49, 50$ is 46. With an even number of numbers, the median is the mean of the two middle numbers: the median of $6, 7, 9, 10$ is $(7 + 9) \div 2 = 8$. Note: when a number is added or changed, the "middle" position can move, so re-order the list.
6. Consecutive integers. For an odd number of consecutive integers, the mean is the middle number: the mean of $19, 20, 21, 22, 23$ is 21.

Why is it true? Why do the deviations add up to 0? Suppose 4 numbers have mean 80, so their sum is $4 \times 80 = 320$. Write each number as 80+ (its deviation). Adding up: 320+ (sum of deviations) $= 320$, so the sum of the deviations must be 0. □



The four numbers $38, 43, 41, 38$ have mean 40. Deviations from 40: -2, $+3$, $+1$, -2; sum $= 0$.

Note for parents. You can illustrate the mean with stacks of coins: "level out" the tall stacks into the short ones until all stacks are the same height. That common height is the mean. The coins moved out of the tall stacks exactly equal the coins added to the short ones – this is the idea "the deviations add up to 0".

Worked examples

Example 1

Consider the set $\{0.5;\ 0.45;\ 0.405;\ 0.054;\ 0.54\}$.

a) Arrange the numbers in increasing order.

b) Find the sum of the largest and the smallest number.

Guided solution.

Step 1. Write the numbers in a column, adding zeros to get three digits after the point

Adding zeros at the end of the decimal part does not change the value, so:

$$0.5 = 0.500; \quad 0.45 = 0.450; \quad 0.405; \quad 0.054; \quad 0.54 = 0.540.$$

Step 2. Compare them like whole numbers

Dropping "0.", we compare 500; 450; 405; 54; 540:

$$54 < 405 < 450 < 500 < 540.$$

So the increasing order is: $0.054 < 0.405 < 0.45 < 0.5 < 0.54$.

Step 3. Add the largest and the smallest (decimal points lined up)

$$\begin{array}{r} 0.540 \\ +\ 0.054 \\ \hline 0.594 \end{array}$$

Check: $0.594 - 0.54 = 0.054$. Correct.

Answer: **a)** $0.054 < 0.405 < 0.45 < 0.5 < 0.54$; **b)** 0.594

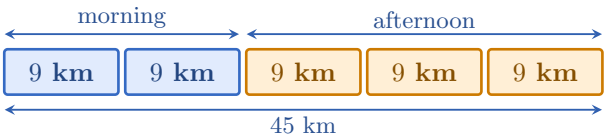
Example 2

Hoa cycles along a 45 km route. In the morning she covers $\frac{2}{5}$ of the route. How many kilometres does she still have to ride in the afternoon, and what fraction of the whole route is that?

Guided solution.

Step 1. Draw a bar model

Split the route into 5 equal parts; the morning covers 2 parts, the afternoon 3 parts.



Step 2. Find the value of one part

One part is $45 \div 5 = 9$ (km).

Step 3. Morning and afternoon distances

Morning: $2 \times 9 = 18$ (km). Afternoon: $45 - 18 = 27$ (km), which is also $3 \times 9 = 27$ (km).

Step 4. **The fraction left** $1 - \frac{2}{5} = \frac{5}{5} - \frac{2}{5} = \frac{3}{5}$. Check: $\frac{27}{45} = \frac{27 \div 9}{45 \div 9} = \frac{3}{5}$.

Answer: 27 **km, which is** $\frac{3}{5}$ **of the route**

Example 3

a) Evaluate $20 - 4 \times 3 + 8 \div 2$ and $(20 - 4) \times (3 + 8) \div 2$.
b) Without calculating exactly, decide which number 101×49 is closest to: 50; 500; 5000; 50 000? Then compute it exactly with a quick method.

Guided solution.

Step 1. **No brackets: multiply and divide first; add and subtract after**

$$20 - \underbrace{4 \times 3}_{12} + \underbrace{8 \div 2}_{4} = 20 - 12 + 4 = 8 + 4 = 12.$$

(Subtraction and addition are done from left to right: $20 - 12 = 8$, then $8 + 4 = 12$.)

Step 2. **With brackets: brackets first**

$$(20 - 4) \times (3 + 8) \div 2 = 16 \times 11 \div 2 = 176 \div 2 = 88.$$

Step 3. **Estimate**
$101 \approx 100$ and $49 \approx 50$, so $101 \times 49 \approx 100 \times 50 = 5000$. Choose 5000.

Step 4. **Exact value using the distributive property**
$101 \times 49 = (100 + 1) \times 49 = 4900 + 49 = 4949$, indeed very close to 5000.

Answer: **a)** 12 **and** 88; **b) closest to** 5000; $101 \times 49 = 4949$

Example 4

a) Rice costs 18 thousand dong per kilogram. Mum buys 15 kg and gets a 10% discount on the whole bill. How much does she pay?
b) A shirt costs 400 thousand dong. After every week the shop reduces the *current price* by 10%. After at least how many weeks is the price of the shirt below 300 thousand dong?

Guided solution.

Step 1. **Total before the discount**
Total cost = unit price × quantity = $18 \times 15 = 270$ (thousand dong).

Step 2. **Subtract** 10%
10% of 270 is $270 \div 10 = 27$ (thousand dong). Mum pays $270 - 27 = 243$ (thousand dong) = 243 000 dong. (Or: 90% remains, i.e. $270 \times 0.9 = 243$.)

Step 3. **Successive discounts: multiply by** 0.9 **every week**

Time	Start	After 1 week	After 2 weeks	After 3 weeks
Price (thousand dong)	400	360	324	291.6
Reduction	–	40	36	32.4

After 2 weeks the price is $324 > 300$; after 3 weeks it is $291.6 < 300$.

Common mistake. Do not think "the price drops by 40 thousand every week" (because 10% of 400 is 40). Each week the reduction is 10% of *the price at that time*, so the reductions get smaller and smaller: 40, then 36, then 32.4, ...

Answer: a) 243 000 dong; b) 3 weeks

Example 5

a) Bình cycles 3 laps around a stadium with a perimeter of 400 m in 4 minutes. An cycles at 15 km/h. Who is faster?
b) A car travels at 72 km/h. How many metres does it travel in 3 seconds?

Guided solution.

Step 1. Bình's speed (m/s)
Distance: $3 \times 400 = 1200$ (m). Time: 4 minutes $= 4 \times 60 = 240$ (seconds).

$$v_{\text{Bình}} = 1200 \div 240 = 5 \text{ (m/s)}.$$

Step 2. Convert An's speed to m/s

Recall. $1 \text{ km/h} = \dfrac{1000 \text{ m}}{3600 \text{ s}} = \dfrac{5}{18} \text{ m/s}.$

$v_{\text{An}} = 15 \times \dfrac{5}{18} = \dfrac{75}{18} \approx 4.17$ (m/s). Since $5 > 4.17$, **Bình is faster**. (Another way: 5 m/s $= 5 \times 3.6 = 18$ km/h > 15 km/h.)

Step 3. The car
72 km/h $= 72 \times \dfrac{5}{18} = 20$ m/s. In 3 seconds it travels $20 \times 3 = 60$ (m).

Answer: a) Bình is faster (5 m/s against about 4.17 m/s); b) 60 m

Example 6

a) Which of the numbers 72; 84; 117; 135 are divisible by 9?
b) Find the digit x such that the number $\overline{3x5}$ is divisible by 9.
c) How many whole numbers are divisors of both 24 and 36?

Guided solution.

Step 1. Use the rule for 9: compute the digit sum
72: $7 + 2 = 9$ (divisible by 9). 84: $8 + 4 = 12$ (not divisible by 9).
117: $1 + 1 + 7 = 9$ (divisible). 135: $1 + 3 + 5 = 9$ (divisible).
So 72, 117, 135 are divisible by 9; 84 is not ($84 = 9 \times 9 + 3$).

Step 2. Find the digit x
The digit sum of $\overline{3x5}$ is $3 + x + 5 = 8 + x$. Since x is a digit, $8 \le 8 + x \le 17$. In this range only 9 is divisible by 9, so $8 + x = 9$, i.e. $x = 1$. The number is $315 = 9 \times 35$.

Step 3. Common divisors of 24 and 36
Divisors of 24: $1, 2, 3, 4, 6, 8, 12, 24$. Divisors of 36: $1, 2, 3, 4, 6, 9, 12, 18, 36$.
The numbers in both lists: $1, 2, 3, 4, 6, 12$ – these are exactly the divisors of $\gcd(24, 36) = 12$.

Answer: a) 72; 117; 135; b) $x = 1$; c) 6 numbers

Example 7

In the calculation $\overline{AB} + \overline{AB} + \overline{AB} = \overline{1AB}$, each letter stands for a digit. Find A and B.

Guided solution.

Method 1 – Place value. We have $\overline{1AB} = 100 + \overline{AB}$. So

$$3 \times \overline{AB} = 100 + \overline{AB} \Rightarrow 2 \times \overline{AB} = 100 \Rightarrow \overline{AB} = 50.$$

So $A = 5$, $B = 0$.

Method 2 – Column by column. Ones column: $B + B + B = 3B$ ends in the digit B, so $3B - B = 2B$ ends in 0, hence $B = 0$ or $B = 5$.

- If $B = 5$: $3 \times 5 = 15$, write 5 carry 1. Tens column: $3A + 1 = \overline{1A} = 10 + A$, so $2A = 9$ – no digit works. Rejected.
- If $B = 0$: no carry. Tens column: $3A = 10 + A$, so $2A = 10$, $A = 5$.

Check: $50 + 50 + 50 = 150 = \overline{1AB}$ with $A = 5$, $B = 0$. Correct.

Answer: $A = 5$, $B = 0$

Example 8

a) Nam's four test scores are 6; 9; 10 and x. The mean of the four scores is 8. Find x.
b) The mean of five consecutive whole numbers is 30. Find the smallest of them.

Guided solution.

Step 1. Using the sum

Recall. Sum = mean × how many numbers.

The four scores add up to $8 \times 4 = 32$. So $x = 32 - 6 - 9 - 10 = 7$.

Step 2. Using deviations (faster)

Compared with 8: the 6 is 2 short, the 9 is 1 over, the 10 is 2 over. Sum of deviations: $-2+1+2 = +1$. For all deviations to add up to 0, the fourth score must be 1 short: $x = 8-1 = 7$.

Step 3. Five consecutive numbers

For five consecutive numbers, the mean is the middle number. The middle number is 30, so the five numbers are $28, 29, 30, 31, 32$. The smallest is 28.

Check: $28 + 29 + 30 + 31 + 32 = 150 = 5 \times 30$. Correct.

Answer: a) $x = 7$; b) 28

Core practice

Problem 1

The smallest number in the set $\{3.2;\ 2.3;\ 3;\ 2.23;\ 3.22\}$ is

(A) 3.2 (B) 2.3 (C) 3 (D) 2.23 (E) 3.22

Your work:

Problem 2

Minh goes on a 36 km kayak trip. Before lunch he paddles the first 12 km. After lunch, what fraction of the whole trip does Minh still have to paddle?

(A) $\frac{1}{2}$ (B) $\frac{5}{6}$ (C) $\frac{3}{4}$ (D) $\frac{2}{3}$ (E) $\frac{3}{5}$

Your work:

Problem 3

Which of the following statements is **true**?

(A) 0 is less than -5 (B) 7 is less than -1 (C) 10 is less than $\frac{1}{4}$

(D) -1 is less than -3 (E) -8 is less than -2

Your work:

Problem 4

The value of $6 \times 2 - 3$ is

(A) 9 (B) -6 (C) 12 (D) 15 (E) 10

Your work:

Problem 5

A test is marked out of 50. Mai scores 80%. How many marks does Mai get?

(A) 40 (B) 62.5 (C) 10 (D) 45 (E) 35

Your work:

Problem 6

Minced beef sells for 120 000 dong per kilogram. How much do 12 kg of minced beef cost?

(A) 120 000 dong (B) 12 000 dong (C) 1 440 000 dong

(D) 132 000 dong (E) 10 000 dong

Your work:

Problem 7

The value of $(8 \times 4) + 3$ is

(A) 96 (B) 15 (C) 56 (D) 35 (E) 28

Your work:

Problem 8

Which of the following integers is closest to 99×9?

(A) 10 000 (B) 100 (C) 100 000 (D) 1000 (E) 10

Your work:

Problem 9

Huy is reading a 396-page book. He has read only the first $\frac{1}{3}$ of the book. How many pages does he still have to read to finish the book?

(A) 264 (B) 124 (C) 250 (D) 199 (E) 244

Your work:

Problem 10

Which of the following numbers is a multiple of 9?

(A) 50 (B) 40 (C) 35 (D) 45 (E) 55

Your work:

Problem 11

Consider the set $\{0.34;\ 0.304;\ 0.034;\ 0.43\}$. The sum of the smallest and the largest numbers in the set is

(A) 0.77 (B) 0.734 (C) 0.077 (D) 0.464 (E) 0.338

Your work:

Problem 12

Lan and Hà run on a circular track 150 m long. Lan runs 3 laps in 120 seconds, and Hà runs 5 laps in 160 seconds. Who is the faster runner, and approximately what is her speed?

(A) Lan, 3.75 m/s (B) Lan, 2.4 m/s (C) Hà, 3.3 m/s

(D) Hà, 4.69 m/s (E) Hà, 3.75 m/s

Your work:

Problem 13

In the addition shown, P and Q each represent a digit. The value of $P + Q$ is

$$\begin{array}{r} P\,Q\,Q \\ P\,P\,Q \\ +\,Q\,Q\,Q \\ \hline 8\;7\;6 \end{array}$$

(A) 3 (B) 5 (C) 7 (D) 6 (E) 4

Your work:

Problem 14

The mean (average) of the four integers 78; 83; 82 and x is 80. Which of the following statements is true?

(A) x is 2 greater than the mean

(B) x is 1 less than the mean

(C) x is 2 less than the mean

(D) x is 3 less than the mean

(E) x is equal to the mean

Your work:

Problem 15

In the Fibonacci sequence $1, 1, 2, 3, 5, \ldots$, each number from the 2 onwards is the sum of the two numbers just before it. For example, the next number in the sequence is $3 + 5 = 8$. Which of the following numbers is in the sequence?

(A) 20 (B) 21 (C) 22 (D) 23 (E) 24

Your work:

Problem 16

The mean (average) of five consecutive integers is 21. The smallest of the five integers is

(A) 17 (B) 21 (C) 1 (D) 18 (E) 19

Your work:

Problem 17

How many positive whole numbers, including 1, divide exactly into both 40 and 72?

(A) 9 (B) 12 (C) 4 (D) 2 (E) 5

Your work:

Problem 18

The number that goes into the box to make $\frac{3}{7} = \frac{\square}{63}$ true is

(A) 27 (B) 9 (C) 59 (D) 63 (E) 3

Your work:

Problem 19

A fraction is equal to $\frac{5}{8}$. Its numerator and denominator add up to 91. What is the difference between the denominator and the numerator of this fraction?

(A) 21 (B) 3 (C) 33 (D) 13 (E) 19

Your work:

Problem 20

Lan is driving at 80 km/h along a road with equally spaced power poles. Her car passes a pole every 4 seconds. Which of the following is closest to the distance between two neighbouring poles?

(A) 50 m (B) 60 m (C) 70 m (D) 80 m (E) 90 m

Your work:

Problem 21

Each of the integers 334 and 419 has digits whose product is 36 (since $3 \cdot 3 \cdot 4 = 36$ and $4 \cdot 1 \cdot 9 = 36$). How many 3-digit positive integers have digits whose product is 36?

(A) 21 (B) 15 (C) 18 (D) 24 (E) 12

Your work:

Problem 22

An *arithmetic sequence* is a sequence in which each term after the first is obtained by adding a constant to the previous term. For example, $2, 4, 6, 8$ and $1, 4, 7, 10$ are arithmetic sequences.
In the grid shown, the numbers in each row (read from left to right) must form an arithmetic sequence, and the numbers in each column (read from top to bottom) must form an arithmetic sequence. The value of x is

1			
4			25
7			x
10		36	

(A) 37 (B) 28 (C) 36 (D) 43.75 (E) 46

Your work:

Problem 23

In the six-digit number $\overline{1ABCDE}$, each letter represents a digit. Given that $\overline{1ABCDE} \times 3 = \overline{ABCDE1}$, the value of $A + B + C + D + E$ is

(A) 29 (B) 26 (C) 22 (D) 30 (E) 28

Your work:

Problem 24

On a coordinate grid, a dot starts at the point $(20, 19)$. On its first move it goes 1 unit horizontally or vertically, that is, to one of the points $(21, 19)$, $(19, 19)$, $(20, 20)$ or $(20, 18)$. On its second move it goes 2 units in a direction perpendicular to the first move (either way). Each move after that is 1 unit longer than the previous one (3 units, 4 units, 5 units,...) and must be perpendicular to the direction of the move just before it. The dot stops after 10 moves. Which of the following final positions is **not** possible?

(A) $(27, 33)$ (B) $(30, 40)$ (C) $(21, 21)$ (D) $(42, 44)$ (E) $(37, 37)$

Your work:

Problem 25

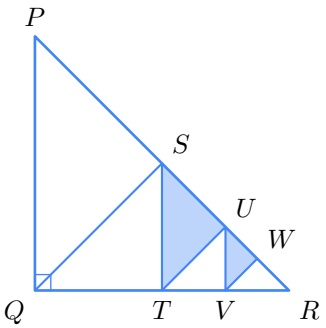
In the "Coin Game", you toss three coins at the same time. You win only if the 3 coins all show heads or all show tails. If you play the game once only, what is the probability of winning?

(A) $\frac{1}{6}$ (B) $\frac{1}{4}$ (C) $\frac{2}{27}$ (D) $\frac{2}{3}$ (E) $\frac{1}{3}$

Your work:

Problem 26

In the right-angled triangle PQR, the right angle is at Q and $PQ = QR$. The segments QS, TU and VW are perpendicular to PR, and the segments ST and UV are perpendicular to QR, as shown. What fraction of the area of triangle PQR is shaded?



(A) $\frac{3}{16}$ (B) $\frac{3}{8}$ (C) $\frac{5}{16}$ (D) $\frac{5}{32}$ (E) $\frac{7}{32}$

Your work:

Problem 27

Mai went to a second-hand market where, every 15 minutes, the price of every item was reduced by 10% of its *current* price. At 9:00 a.m., the price of a rug was 100 000 dong. At 9:15 a.m., the price was reduced to 90 000 dong. As soon as the price of the rug fell *below* 80 000 dong, Mai bought it. At what time did Mai buy the rug?

(A) 9:45 a.m. (B) 9:15 a.m. (C) 9:30 a.m.

(D) 10:15 a.m. (E) 10:00 a.m.

Your work:

Problem 28

The number 2018 is used to create six-digit positive integers: each of them must contain the digits $2, 0, 1, 8$ *together and in this order*. For example, 720 186 is allowed, but 209 318 and 210 893 are not. How many of these six-digit integers are divisible by 9?

(A) 28 (B) 27 (C) 31 (D) 34 (E) 22

Your work:

Problem 29

The number N is the product of all positive odd integers from 1 to 99 that do *not* end in the digit 5. That is,

$$N = 1 \times 3 \times 7 \times 9 \times 11 \times 13 \times 17 \times 19 \times \cdots \times 91 \times 93 \times 97 \times 99.$$

The units digit of N is

(A) 1 (B) 3 (C) 5 (D) 7 (E) 9

Your work:

Problem 30

Five students wrote a quiz with a maximum score of 50. The scores of four of the students were 42, 43, 46 and 49. The score of the fifth student was N. The mean of the five students' scores was the same as the median of the five students' scores. The number of possible values of N is

(A) 3 (B) 4 (C) 1 (D) 0 (E) 2

Your work:

Extension and challenge

Problem 31

Consider the numbers: -1.5; 0.25; -0.75; 1.2; -2.

a) Arrange the numbers in increasing order.

b) Which numbers lie between -1 and 1 on the number line?

c) Find the sum of the largest and the smallest number.

Your work:

Problem 32

Consider the expression $2 + 3 \times 4 - 1$.

a) Evaluate the expression.

b) Insert *one* pair of brackets ( ) into the expression (without changing the order of the numbers or the signs) to get the largest possible value. What is that largest value?

c) Same question as b), but for the smallest value.

Your work:

Problem 33

An's storybook has 360 pages. On the first day An reads $\frac{1}{4}$ of the book. On the second day An reads $\frac{1}{3}$ *of the remaining pages*. After two days, how many pages has An not read yet, and what fraction of the book is that?

Your work:

Problem 34

The first test has 40 questions, and Bình answers 32 of them correctly.

a) What percentage of the questions in the first test did Bình answer correctly?

b) The second test has 60 questions. At least how many questions must Bình answer correctly in the second test so that, over both tests together, his correct answers make up at least 85% of all the questions?

Your work:

Problem 35

A shop sells cashew nuts in packs: a 3 kg pack costs 255 thousand dong, a 5 kg pack costs 410 thousand dong (no loose sales).

a) Which pack is cheaper per kilogram?

b) Ms Hằng needs to buy *exactly* 15 kg. How should she buy them to spend as little as possible? How much does it cost?

c) How much does it cost to buy *exactly* 13 kg?

Your work:

Problem 36

The five members of a group weigh (in kg) 38; 41; 40; 37 and x. Their mean weight is 40 kg.

a) Find x. Is the fifth member heavier or lighter than the mean, and by how many kilograms?

b) Find the median weight of the group.

c) A sixth member joins the group, and the mean weight of all six becomes 41 kg. How much

does the sixth member weigh?

Your work:

Problem 37 *

A running track around the schoolyard is 200 m long. Mai runs 6 laps in 4 minutes. Tú runs at a constant 18 km/h.

a) Who is faster?

b) Keeping the same speeds, how many minutes does each of them need to run 3 km?

c) They start at the same time from the same point of the track, but run in *opposite directions*. After how many seconds do they meet for the first time?

d) One day Tú increases his speed to 21.6 km/h while Mai runs as before. They start at the same time from the same point, running in the *same direction*. How long is it until Tú is exactly one lap ahead of Mai?

Your work:

Problem 38

a) Find the digits a, b such that the number $\overline{4a7b}$ is divisible by 2, 5 and 9.

b) How many four-digit numbers of the form $\overline{5a3b}$ are divisible by 9? List them.

Your work:

Problem 39

In the addition below, M and N are two different non-zero digits. Find all pairs $(M; N)$ that work.

$$\begin{array}{r} M\ N \\ N\ M \\ +\ N\ N \\ \hline 1\ 8\ 7 \end{array}$$

Your work:

Problem 40 **

A sequence is built by the Fibonacci rule: from the third number on, each number is the sum of the two numbers just before it.

a) The first number is 2 and the fifth number is 19. Find the second and the eighth numbers of the sequence.

b) Find the mean of the first five numbers of the sequence in part a).

c) In the familiar Fibonacci sequence $1, 1, 2, 3, 5, 8, \ldots$, is the 2026th number even or odd? Explain.

Your work:

Problem 41 *

Sums of consecutive numbers (extends Problem 16).

a) Find *all* the ways of writing 100 as a sum of at least two consecutive positive integers.

b) Prove that 64 cannot be written as a sum of at least two consecutive positive integers.

Your work:

Problem 42

Common divisors and counting divisors (extends Problem 17).

a) How many positive integers divide both 720 and 1080?

b) Find the smallest positive integer with exactly 12 divisors.

Your work:

Problem 43

How does a fraction change? (extends Problems 18 and 19).

a) When the same whole number n is added to both the numerator and the denominator of $\frac{5}{8}$, the result is a fraction equal to $\frac{4}{5}$. Find n.

b) Let $\frac{a}{b}$ be a fraction with $0 < a < b$. Prove that for every positive integer n, $\frac{a+n}{b+n} > \frac{a}{b}$.

c) How many fractions equal to $\frac{3}{7}$ have a two-digit denominator?

Your work:

Problem 44 *

Poles, a bicycle and a train (extends Problem 20).

a) An cycles steadily at 18 km/h along a road whose power poles are equally spaced 50 m apart. How many minutes pass from the moment An passes the first pole to the moment he passes the 25th pole?

b) A train travelling at constant speed takes 9 seconds to pass a power pole and 30 seconds to cross a bridge 350 m long (from the moment the front of the train reaches the bridge until the end of the train has just left it). Find the length of the train and its speed in km/h.

Your work:

Problem 45 **

Products of digits (extends Problem 21).

a) How many four-digit numbers have digits whose product is 36?

b) Find the smallest positive integer whose digits have product 360.

c) Find the largest positive integer with *all different* digits whose product is 360.

Your work:

Problem 46 ★

Arithmetic grids (extends Problem 22). In a square grid, the numbers in every row and in every column form arithmetic sequences.

a) A 4×4 grid has these four corner numbers: top left 1, top right 10, bottom left 19, bottom right 37. Fill in the whole grid.

b) Find the sum of all 16 numbers in the grid of part a). Compare it with the sum of the four corners: what do you notice? Explain.

c) Prove that in a 3×3 grid of this kind, the number in the centre always equals the mean of the four corner numbers.

1			10
19			37

Your work:

Problem 47 ⋆⋆

"Rotating" numbers (extends Problem 23). In each part, every letter represents a digit, and $A \neq 0$.

a) Find the six-digit number $\overline{2ABCDE}$ such that $\overline{2ABCDE} \times 3 = \overline{ABCDE2}$.

b) Find all digits d (from 1 to 9) for which there is a six-digit number $\overline{dABCDE}$ with $\overline{dABCDE} \times 3 = \overline{ABCDEd}$.

c) Prove that there is no six-digit number $\overline{1ABCDE}$ with $\overline{1ABCDE} \times 2 = \overline{ABCDE1}$.

Your work:

Problem 48 ⋆⋆

A frog on the number line (extends Problem 24). A frog sits at the point 0 on the number line. On its k-th jump ($k = 1, 2, 3, \dots$), the frog jumps exactly k units, to the left or to the right as it likes.

a) After exactly 10 jumps, can the frog be back at 0? Why?

b) After exactly 10 jumps, which points can the frog be at? How many such points are there?

c) What is the smallest number of jumps the frog needs to reach the point 20?

Your work:

Problem 49 *

Numbers containing the block 2025 (extends Problem 28). Consider the six-digit numbers that contain the four digits $2, 0, 2, 5$ *together and in this order* (for example: 720 251, 202 513, 132 025). How many of these numbers are divisible by 45?

Your work:

Problem 50 **

Mean and median (extends Problem 30). A list of five positive integers (repeats allowed) has mean 12 and median 10.

a) What is the largest possible value of the largest number in the list?

b) What is the smallest possible value of the largest number in the list?

c) Answer both questions again if the five numbers must all be different.

Your work: